

Pseudocode for MATLAB Code: Optimized RAM Structure using Bayesian Optimization

1. ****Initialization and Setup****

- ****Define frequency range****:
 - ``f = 0.1:0.01:1`` GHz (from 0.1 to 1 GHz in steps of 0.01 GHz)
 - ``numFrequencies = length(f)`` (number of frequencies)
 - Convert frequency to Hz: ``f_Hz = f * 1e9``
- ****Set the speed of light****:
 - ``c = 3e8`` m/s

2. ****Material Property Initialization****

- ****Initialize permittivity (epsr)****:
 - ``M_epsr = zeros(16, numFrequencies)``
 - Set constant values for materials 1-5 and 9-16.
 - Frequency-dependent permittivity for materials 6-8 using the given formulas.
- ****Initialize permeability (mur)****:
 - ``M_mur = zeros(16, numFrequencies)``
 - Set constant permeability values for materials 1-2 and 6-8.
 - Frequency-dependent permeability for materials 3-5 and 9-16 using given formulas.

3. ****Bayesian Optimization Setup****

- ****Define the number of layers****: ``nVar = 3`` (three layers in the structure)
- ****Define optimization variables****:
 - For each layer, define thickness and material choice as optimization variables:
 - ``thicknessVars(i) = optimizableVariable('Thickness', [0.1, 7.0], 'Type', 'real')``
 - ``materialVars(i) = optimizableVariable('Material', [1, 16], 'Type', 'integer')``
 - Store all variables in ``vars = [thicknessVars, materialVars]``.
- ****Objective function setup****:
 - Define the function to be optimized: ``fun = @(params) reflectionObjective_BO(params, f, M_epsr, M_mur)``.

4. ****Running Bayesian Optimization****

- ****Perform Bayesian optimization****:
 - ``results = bayesopt(fun, vars, ...)``
 - Set optimization parameters:
 - ``MaxObjectiveEvaluations = 20`` (increase the number of evaluations)

- `NumSeedPoints = 50` (increase initial exploration points)
- `AcquisitionFunctionName = 'expected-improvement'`
- `ExplorationRatio = 3` (increase exploration for variability)
- Plot results during optimization: `PlotFcn = {@plotMinObjective, @plotAcquisitionFunction}`.
- `Verbose = 1` (display optimization progress).

5. ****Extracting Optimal Parameters****

- ****Extract optimal thickness and material selections****:
 - `opt\_params = results.XAtMinObjective`
 - `opt\_thickness = [opt\_params.Thickness1, opt\_params.Thickness2, opt\_params.Thickness3]`
 - `opt\_material\_selection = [opt\_params.Material1, opt\_params.Material2, opt\_params.Material3]`
- ****Display results****:
 - Output the optimal thickness, material selection, and the minimum reflection achieved:
 - `disp('Optimal Thickness (mm):')`
 - `disp(opt\_thickness)`
 - `disp('Optimal Material Selection:')`
 - `disp(opt\_material\_selection)`
 - `disp('Minimum Reflection in dB:')`
 - `disp(results.MinObjective)`

6. ****Reflection Calculation Using Optimal Parameters****

- ****Initialize permittivity (epsr) and permeability (mur)**** for the selected materials:
 - `epsr = [ones(1, numFrequencies); M\_epsr(mat\_selection, :); ones(1, numFrequencies)]`
 - `mur = [ones(1, numFrequencies); M\_mur(mat\_selection, :); ones(1, numFrequencies)]`
- ****Calculate reflection coefficients for each frequency****:
 - Loop over the frequency range:
 - `for i = 1:numFrequencies`
 - Compute reflection using `RMultiSlab` function:
 - `[rSlab\_TE\_abs, rSlab\_TM\_abs] = RMultiSlab(nVar, 0, epsr(:,i), mur(:,i), f(i), thickness)`
 - Store TE reflection coefficients in `rSlab\_TE\_abs1(i, :)`.
 - Calculate the TE reflection in dB:
 - `a(i) = rSlab\_TE\_abs1(i, nVar+1)`
 - `rSlab\_TE\_db(i) = 10 \* log10(a(i))`

7. ****Plot the Optimized Reflection****

- ****Plot the optimized reflection****:
 - Use ``semilogx`` to plot reflection in dB:
 - ``semilogx(f, rSlab_TE_db, 'r', 'LineWidth', 2.5)``
 - Set title and axis labels:
 - ``title('Optimized RAM Structure')``
 - ``xlabel('Frequency (GHz)')``
 - ``ylabel('Max Reflection in dB')``
 - ``grid on``

8. ****Reflection Calculation Objective Function for Bayesian Optimization****

- ****Define the objective function****:
 - The objective function calculates the maximum reflection in dB over the frequency range.
 - Inputs: Thickness and material choices.
 - Compute permittivity (``epsr``) and permeability (``mur``) for selected materials.
 - Use ``RMultiSlab`` to calculate reflection coefficients for each frequency.
 - Convert the reflection to dB and return the maximum reflection as the objective value.
 - ``reflection_db = max(rSlab_TE_db)``.